

Juniper Networks® vMX Release Notes

Release 17.3R1
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These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

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Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 17.3 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

This release of vMX supports most of the features available on Juniper Networks MX Series routers with the following exception:

- High availability features are not supported in this release.

New and Changed Features

This section describes the new features and enhancements in this release.

- **Class of Service support for four-level hierarchical scheduling**—Four-level hierarchical scheduling is supported for 16 level 2 nodes. You must set the maximum number of hierarchical scheduling levels to 3 by configuring the **maximum-hierarchy-levels 3** option at the `[edit interfaces interface-name hierarchical-scheduler]` hierarchy level. If the **maximum-hierarchy-levels** option is not configured, it is automatically set to 2. You set the number of level 2 nodes by configuring the **maximum-l2-nodes *number-of-nodes*** option at the `[edit interfaces interface-name hierarchical-scheduler]` hierarchy level where *number-of-nodes* is the number of level 2 nodes; the default value is 4 and two of those nodes are used for control traffic. If you configure less than four nodes, the value is accepted but there are not enough nodes for other applications to use. You enable CoS by configuring the **flexible-queuing-mode** option at the `[edit chassis fpc 0]` hierarchy level.



NOTE: The FPC reboots if you change the number of level 2 nodes or enable CoS.

[See [Understanding Hierarchical CoS for Subscriber Interfaces](#).]

- **Group VPN support**—vMX supports Group VPNs. You enable inline service interfaces by configuring the **inline-services bandwidth (1g | 10g)** option at the `[edit chassis fpc 0 pic 0]` hierarchy level. The bandwidth value is not used for si traffic, so you can choose either value.



NOTE: The FPC reboots if you enable inline service interfaces.

vMX supports the same Group VPNv2 features as MX Series routers. Group VPNv2 is the name of the Group VPN technology on MX Series routers. Group VPNv2 integrates routing and encryption in the network. vMX does not support match-direction input configuration for interface-style configuration. vMX does not support fail-open, partial fail-open, or time-based anti-replay protection.

[See [Configuring Group VPNs](#).]

- **vMX mode display**—By default, performance mode is enabled for vMX. If you have not configured enough vCPUs, vMX runs in lite mode. To display the mode in which vMX is running (performance mode or lite mode), the output of the **show chassis hardware** command displays the mode in the part number field associated with the FPC. **RIOT-PERF** indicates performance mode and **RIOT-LITE** indicates lite mode.

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).
- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.
- ICMP echo request packets are handled inline, which means ICMP packets do not reach the VCP and are replied to by the VFP itself. No ICMP packets are seen on the VCP and packet capture tools do not capture ICMP packets on the VCP.

Known Issues

This section lists the known issues in this release.

- When the FPC is restarted, **kernel: GENCFG: op 32 (Resync blob) failed** syslog message appears. This harmless message can be ignored. PR1050467
- LACP packets are getting dropped on the bridge. PR1059231
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- When committing the configuration, **%DAEMON-4-SNMP_EVLIB_FAILURE: PFED ran out of transfer credits with PFE.Failed to get stats.** messages sometimes appear in the syslog file. PR1124902

- Multiple vMX instances using SR-IOV on the same host are not supported. PR1130534
- When committing the configuration, **pfe: %USER-3: downward spike received from pfe for opackets_reply:3545 opackets_record:48932521** messages sometimes appear in the syslog file. PR1146002
- Many **clksync_update_tod_to_pfe:109 Failed to send non-PEC pfe TOD update to other PFEs. Error code: 29.** messages appear in the syslog file. PR1148344
- Scheduler slips are occurring in large VPLS configurations and displaying **JTASK_SCHED_SLIP** log messages. PR1153290
- Observing **rp[7116]: %DAEMON-6: ifl_delete: ifl** error messages that indicate a multicast tunnel (mt) has been deleted. PR1156725
- CPU pinning is not working properly with OVS bridges. PR1171096

Subscriber Management

- When bringing up an LNS session on vMX, the Cos-Shaping-Rate attribute (ERX-Attr-177) is sometimes omitted from the Acct-Start messages sent to RADIUS. PR1167154

System Requirements

These topics provide system requirements for each supported environment.

- [Minimum Hardware and Software Requirements for KVM on page 4](#)
- [Minimum Hardware and Software Requirements for VMware on page 7](#)

Minimum Hardware and Software Requirements for KVM

[Table 1 on page 4](#) lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.

Table 1: Minimum Hardware Requirements for KVM (continued)

Description	Value
Number of cores NOTE: Performance mode is the default mode and the minimum value is based on one port. The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the formula to calculate the minimum vCPUs needed by the VFP. To calculate the optimal number of vCPUs needed by VFP for performance mode: To calculate the optimal number of vCPUs needed by VFP for performance mode: - Without CoS—$(4 * \text{number-of-ports}) + 4$ - With CoS—$(5 * \text{number-of-ports}) + 4$ All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.	For lab simulation use case applications (lite mode): Minimum of 4 1 for VCP 3 for VFP For low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications (performance mode): Minimum of 9 1 for VCP 8 for VFP
Memory NOTE: Performance mode is the default mode.	For lab simulation use case applications (lite mode): Minimum of 5 GB 1 GB for VCP 4 GB for VFP For low-bandwidth or high-bandwidth applications (performance mode): Minimum of 16 GB 4 GB for VCP 12 GB for VFP Additional 2 GB recommended for host OS
Storage	Local or NAS
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

Table 2 on page 6 lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 14.04.1 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages NOTE: Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev NOTE: libvirt 1.2.19

[Table 3 on page 6](#) lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5
Virtualization	QEMU-KVM 1.5.3
Required packages NOTE: SR-IOV requires these packages: kernel-devel gcc	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools NOTE: libvirt 1.2.17 or later

[Table 4 on page 6](#) lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools NOTE: libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 7 lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

Description	Value
Number of cores NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode: - Without CoS—$(4 * \text{number-of-ports}) + 4$ - With CoS—$(5 * \text{number-of-ports}) + 4$ NOTE: The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.	For performance mode: Minimum of 9 1 for VCP 7 for VFP For lite mode: Minimum of 4 1 for VCP 3 for VFP
Memory NOTE: Performance mode is the default mode.	For performance mode: Minimum of 16 GB 4 GB for VCP 12 GB for VFP For lite mode: Minimum of 10 2 GB for VCP 8 GB for VFP
Storage	Local or NAS

Table 6 on page 7 lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Verifying Security Signatures

The vMX image is securely signed, so you can verify the signature for the image.

To verify the signature:

1. Download the public key certificate and the software image from the Juniper Networks [Download Software](#) page.
2. Import the public key into a temporary GPG key ring.

```
mkdir temp
cd temp
gpg --homedir . --import certificate-file
```

For example:

```
gpg --homedir . --import Juniper_vMX_public_key_2017.asc
gpg: WARNING: unsafe permissions on homedir `.`
Warning: using insecure memory!
gpg: keyring `./secring.gpg' created
gpg: keyring `./pubring.gpg' created
gpg: ./trustdb.gpg: trustdb created
gpg: key CA6E E4DD E89A EB4C F22F 6897 7B82 9893 BA75 0B9B: public key "vMX
Trusted Kernel 2017 <ca@juniper.net>" imported
gpg: Total number processed: 1
gpg:             imported: 1 (RSA: 1)
```



NOTE: The hexadecimal key value and name vary with each key. The hexadecimal value is an easy way to confirm that the key is authentic.

3. Verify the validity of the signature.

```
gpg --homedir . --verify signature-file file-to-be-signed
```

For example:

```
gpg --homedir . --verify vmx-17.2R1.4-1.tgz.sig jinstall-vmx-17.2R1.4-domestic-signed.tgz
gpg: WARNING: unsafe permissions on homedir `.`
Warning: using insecure memory!
gpg: Signature made Tue Jun  6 16:50:05 2017 PDT using RSA key ID BA750B9B
gpg: Good signature from "vMX Trusted Kernel 2017 <ca@juniper.net>"
gpg: This key is certified with a trusted signature!
gpg: WARNING: This key is not certified with a trusted signature!
gpg:             There is no indication that the signature belongs to the owner.
Primary key fingerprint: CA6E E4DD E89A EB4C F22F 6897 7B82 9893 BA75 0B9B
```

The second warning message appears because the key has not been marked as trusted. You can ignore the warning messages.

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at <https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf>.
- Product warranties—For product warranty information, visit <http://www.juniper.net/support/warranty/>.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: <https://www.juniper.net/customers/support/>
- Search for known bugs: <https://prsearch.juniper.net/>
- Find product documentation: <https://www.juniper.net/documentation/>
- Find solutions and answer questions using our Knowledge Base: <https://kb.juniper.net/>
- Download the latest versions of software and review release notes: <https://www.juniper.net/customers/csc/software/>
- Search technical bulletins for relevant hardware and software notifications: <https://kb.juniper.net/InfoCenter/>
- Join and participate in the Juniper Networks Community Forum: <https://www.juniper.net/company/communities/>
- Create a service request online: <https://myjuniper.juniper.net>

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: <https://entitlementsearch.juniper.net/entitlementsearch/>

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit <https://myjuniper.juniper.net>.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see <https://support.juniper.net/support/requesting-support/>.

Revision History

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